

TAEHWA YOO

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Ph.D. student / University of Florida

Bryant Space Science Center, 1772 Stadium Road, Gainesville, FL 32611

EDUCATION

- **Ph.D. student in Astronomy** 2022 - present
University of Florida, US
Advisor: Prof. Adam Ginsburg
- **M.S. in Astronomy** 2018 - 2020
Yonsei University, Republic of Korea
Advisor: Prof. Taysun Kimm
- **B.S. in Astronomy** 2011 - 2018
Yonsei University, Republic of Korea

RESEARCH EXPERIENCE

- **Doctoral research in star formation in high-mass protoclusters, University of Florida** 2022 - present
- Continuum image processing of ALMA long-baseline data / ALMA continuum source extraction / ALMA compact source photometry tool development (TGIF; Two d Gaussian In Fitting)
- JWST NIRCam/MIRI imaging reduction / PSF photometry of JWST compact sources
- **Master's research in high-redshift galaxy simulation, Yonsei University** 2018 - 2020
- Master thesis: "On the origin of low escape fractions from massive star-forming galaxies at high redshift"
- Running and processing of radiation-hydrodynamic simulations using an adaptive mesh refinement code RAMSES-RT
- Experience of using supercomputer systems (Nurion in Korea Institute of Science and Technology Information, JUWELS in Jülich Supercomputing Centre, DiRAC in University of Leicester)

PROPOSALS AND GRANTS

- **JWST: Cycle 3: GO 6151 (PI): *The determination of the upper initial mass function in the nearest massive star-forming region, W51A*** 2024
- The Korean Government Scholarship Program for Study Overseas (\$76,000 for two years) 2022-2023

PUBLICATIONS

- [1] *A JWST NIRCam/MIRI View of the W51A High-mass Star-forming Region* - **Taehwa Yoo**, Adam Ginsburg, Nazar Budaiev, Roberto Galván-Madrid, Aden Dawson, Savannah Gramze, Jesús Hernández, Alexandre Roman-Lopes, Carlos G. Román-Zúñiga, Joel Sanchez-Bermudez, Miriam G. Santa-Maria, Aida Wofford, and Jason E. Ybarra, 2026, AJ, 171, 208
- [2] *ALMA-IMF XX: Core fragmentation in the W51 high-mass star-forming region* - **T. Yoo**, A. Ginsburg, J. Braine, N. Budaiev, F. Louvet, F. Motte, A. M. Stutz, B. Thomasson, M. Armante, M. Bonfand, S. Bontemps, L. Bronfman, G. Busquet, T. Csengeri, N. Cunningham, J. Di Francesco, D. J. Díaz-González, M. Fernández-Lopez, R. Galván-Madrid, C. Goddi, A. Gusdorf, N. Kessler, A. Koley, H.-L. Liu, T. Nony, F. Olguin, P. Sanhueza, M. Valeille-Manet, L. A. Zapata, and Q. Zhang, 2025, ApJ, 994, 233
- [3] *On the origin of low escape fraction of ionizing photons from massive star-forming galaxies at high redshift* - **Taehwa Yoo**, Taysun Kimm, and Joakim Rosdahl, 2020, MNRAS, 499, 5175

- [4] *ALMA-IMF XXII. Role of core subfragmentation in the IMF origin: Hierarchical fragmentation cascade and CMF in W43-MM1* - F. Motte, N. Le Nestour, R. Veyry, N. Brouillet, T. Nony, B. Thomasson, F. Louvet, I. Joncour, E. Moraux, A. Men'shchikov, **T. Yoo**, A. Ginsburg, A. Gusdorf, A. M. Stutz, R. Galván-Madrid, T. Csengeri, R. H. Álvarez-Gutiérrez, M. Armante, Y. Bernard, M. Bonfand, S. Chevalier, N. Cunningham, P. Dell'Ova, M. González, A. Koley, F. A. Olguin, D. Panda, Y. Pouteau, J. Salinas, P. Sanhueza, N. A. Sandoval-Garrido, and M. Valeille-Manet, 2026, arXiv, 2604.14875
- [5] *The coupled origin of the stellar initial mass function and multiplicity: The influence of hierarchical fragmentation on the core mass function* - B. Thomasson, I. Joncour, E. Moraux, F. Motte, **T. Yoo**, and A. Ginsburg, 2026, A&A, 709, A61
- [6] *JWST's First View of the Most Vigorously Star-forming Cloud in the Galactic Center: Sagittarius B2* - Nazar Budaiev, Adam Ginsburg, Ashley T Barnes, Desmond Jeff, **Taehwa Yoo**, Cara Battersby, Alyssa Bulatek, Savannah Gramze, Xing Lu, Elisabeth AC Mills, Theo Richardson, Daniel L Walker, 2026, ApJ, 172, 51
- [7] *Protostellar cores in Sagittarius B2 N and M* - Nazar Budaiev, Adam Ginsburg, Desmond Jeff, Ciriaco Goddi, Fanyi Meng, Álvaro Sánchez-Monge, Peter Schilke, Anika Schmiedeke, **Taehwa Yoo**, 2024, ApJ, 961, 4
- [8] *ACES VII. Compact Continuum Source Catalog of the Central Molecular Zone* - Jennifer Wallace, Cara Battersby, Nazar Budaiev, H Perry Hatchfield, Ashley T Barnes, **Taehwa Yoo**, Miriam G Santa-Maria, Antonio Daley, Q Daniel Wang, Anika Schmiedeke, Kaitlyn E Sheriff, Jairo Armijos-Abendaño, Daniel Walker, Gwenllian Williams, Dani R Lipman, Farideh Mazoochi, Natalie O Butterfield, Francisco Nogueras-Lara, Yoshiaki Sofue, Ralf S Klessen, Simon CO Glover, Katharina Immer, Savannah R Gramze, Katarzyna M Dutkowska, Rojita Buddhacharya, Laura Colzi, Ashley Lieber, Alyssa Bulatek, Michael Wieber, Pablo García, Denise Riquelme-Vásquez, Emad Alkhuja, Yue Hu, Dennis Perlot, Howard A Smith, Victor M Rivilla, Wenyu Jiao, Fengwei Xu, Jonathan D Henshaw, Adam Ginsburg, Elisabeth AC Mills, Qizhou Zhang, Steven N Longmore, Nguyen M Khang, Galaxy Salo, Jakub Poznanski, Dylan Pare, Maya Petkova, Izaskun Jiménez-Serra, Xing Lu, Sergio Martin, Alvaro Sanchez-Monge, Bijas Najimudeen, Christoph Federrath, Tabassum S Tanvir, Pei-Ying Hsieh, John Bally, Paul Ho, Mattia C Sormani, Robin G Tress, Jaime E Pineda, 2026, arXiv, 2069.00135
- [9] *ALMA-IMF: XVII. Census and lifetime of high-mass prestellar cores in 14 massive protoclusters* - M. Valeille-Manet, S. Bontemps, T. Csengeri, T. Nony, F. Motte, A. M. Stutz, A. Gusdorf, A. Ginsburg, R. Galván-Madrid, P. Sanhueza, M. Bonfand, N. Brouillet, P. Dell'Ova, F. Louvet, N. Cunningham, M. Fernández-López, F. Herpin, F. Wyrowski, R. H. Álvarez-Gutiérrez, M. Armante, A. E. Guzmán, N. Kessler, A. Koley, J. Salinas, **T. Yoo**, L. Bronfman, N. Le Nesto, 2025, A&A, 696, A11
- [10] *ALMA-IMF: XVI. Mass-averaged temperature of cores and protostellar luminosities in the ALMA-IMF protoclusters* - F. Motte, Y. Pouteau, T. Nony, P. Dell'Ova, A. Gusdorf, N. Brouillet, A. M. Stutz, S. Bontemps, A. Ginsburg, T. Csengeri, A. Men'shchikov, M. Valeille-Manet, F. Louvet, M. Bonfand, R. Galván-Madrid, R. H. Álvarez-Gutiérrez, M. Armante, L. Bronfman, H-R. V. Chen, N. Cunningham, D. Díaz-González, P. Didelon, M. Fernández-Lopez, F. Herpin, N. Kessler, A. Koley, B. Lefloch, N. Le Nestour, H-L. Liu, E. Moraux, Q. Nguyen Luong, F. Olguin, J. Salinas, N. A. Sandoval-Garrido, P. Sanhueza, R. Veyry, and **T. Yoo**, 2025, A&A, 694, A24

PRESS RELEASE

- *Researchers use James Webb Telescope to reveal hidden details of W51 star formation*, Mar 2026, UF press release

PROGRAMMING PROJECTS

- TGIF (<https://github.com/tyoo37/TGIF>): T. Yoo, & A. Ginsburg. 2024, TGIF: Two d Gaussian In Fitting, v1, Zenodo, doi:10.5281/zenodo.13973837

TECHNICAL SKILLS

- Programming languages: Python (proficient), IDL (intermediate) and Fortran (intermediate)
- Science packages: RAMSES-RT, CASA, CARTA

RESEARCH INTERESTS

- High-mass star formation and the origin of the stellar initial mass function
- Molecular cloud core fragmentation and the connection between the core mass function and the IMF
- Formation and early evolution of massive protostars, including accretion, disks, and outflows
- Multiwavelength observations of star-forming regions with ALMA, JWST, and other facilities

AWARDS

- Excellent Poster Award for Korea Astronomy Society Meeting Oct. 2018, Oct. 2019
- Honors at Yonsei University Feb. 2018
- Korean National Scholarship For Science and Engineering Mar. 2011 - Feb. 2018

ORAL PRESENTATIONS

- *ALMA/JWST view of W51-E and W51-IRS2 protoclusters*, Jun 2026, CMF2IMF workshop
- *Core fragmentation in the W51 high-mass star-forming region*, Sep 2025, Stellar Origins 2025
- *On the origin of low escape fractions of ionizing photons from star-forming galaxies at high redshift*, Dec 2021, SAZERAC SIPS Early Galaxy Formation Near and Far

TEACHING EXPERIENCE

- **Mentoring** - REU program mentoring Avery Lacon with project "SED analysis of the ALMA-JWST matched sources in W51" Summer 2026
- **Teaching Assistant**
 - Galaxies and Universe (1), Yonsei University Fall 2021
 - Observational technique (1), Yonsei University Spring 2021
 - Understanding the Universe, Yonsei University Fall 2019
 - Introduction to Astrophysics (1), Yonsei University Spring 2019
 - Understanding the Universe, Yonsei University Fall 2018
 - Understanding the Universe, Yonsei University Spring 2018

PUBLIC OUTREACH

- Online public lecture on astronomy presented by Yonsei Observatory 2021
- Participation in the production of documentary "Kiss the Universe" by Korea Broadcasting System 2020
- Military Service for Republic of Korea Army 2014 - 2016
- A founding member of public astronomy magazine club Wouldyoulike 2012 - 2016